Jonathan Bong Lin Choon

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Professional Summary

Over 16 years as an I.T professional in the solutions provider, telecommunication, system integration, data-collection, lottery, digital advertising and energy industries; specialising in the areas of software development, testing and documentation, operations and end-user support. Developed new software systems, enhanced existing systems for the education, public sector, airline, financial, lottery (gaming), digital advertising and energy industries. Bachelor of Computing with a double major in systems development and business systems.

Skills

Core Ecosystem: Microsoft .NET | C#

Web Frameworks: ASP.NET | ASP.NET MVC

API & Integration: ASP.NET Web API | RESTful APIs | JSON | XML | XSD

Databases: Microsoft SQL Server | MySQL | SQLite

Tools & Cloud: Html Agility Pack | Selenium Web Driver | Git/GitHub | Sandcastle |
Microsoft Dynamics 365 Finance and Operations |
Microsoft Azure (Vision in Foundry Tools, Data Explorer, SQL, SWA, Storage) |
Amazon Web Services (EC2, S3, RDS, DMS)

Professional Experience

Application Operations Specialist | Sembcorp Utilities Services Pte Ltd (Sembcorp Industries Ltd) | 2018 – 2026

Contributed to the organisation's digital transformation journey for its systems and applications related to energy billing and renewables (solar).

- Spearheaded the digitization and modernisation of legacy billing processes and applications to allow the billing system to work together with interacting systems to meet billing timelines.
- Collaborated with cross-functional teams and external payment vendors to streamline payment processing and improve stakeholder experience.
- In charge of the daily operations and maintenance of the billing system, ensuring its optimal performance and availability, including in disaster recovery scenarios.
- Responsible for the modernisation, enhancement and maintenance of the Solar irradiance applications operating on Solar commercial and industrial sites, regularly collaborating with the Energy Market Authority on site testing and operations, as well as internal stakeholders for downstream solar billing processes.

Software Developer | Smartidea Pte Ltd | 2014 – 2018

Involved in R & D (including proof-of-concept), development, operations, support and maintenance of new/existing systems related to digital advertising.

- Worked closely with the strategic-planning teams and created a software platform for internal sales and business teams to keep track of gaming related and operational statistics to help them develop effective strategies and business proposals to clients.
- Successfully designed and implemented a company-wide payment library system that interacted with payment gateways to the company's merchant banks to make successful payment-related operations.
- Participated in the company's research modules, often consisting of proof-of-concept or investigative work to allow management and stakeholders to conduct and complete feasibility study before deciding to embark on keystone projects.
- Responsible for operations, maintenance and support of the company's SEO platform, ensuring all backend processes are kept active running within normal operational parameters.

Earlier Professional History

- **Applications Analyst | Singapore Pools (Private) Ltd | 2012 – 2014**
- **Consultant, Application | NCS Group | 2009 – 2011**
- **Application Specialist | Singapore Computer Systems (SCS) Ltd (NCS Group) | 2008**
- **Software Developer | Alcatel-Lucent Enterprise - Singapore ALE Pte. Ltd. | 2006 – 2007**
- **Software Engineer | 2005**
- **Programmer | SATO Asia Pacific Pte. Ltd. | 2002**
- **Programmer | Beca | 2000 – 2001**

Open Source & Personal Projects

MSSL EBT System Data Message Model and Interface Definitions | 2026
://jon-bong.github.io/mssl-ebt-models/
C#, XSD, XML Serialisation, .NET Standard, .NET Core, NuGet, Sandcastle, GitHub

An open-source energy market developer class library distributed via NuGet to streamline B2B integrations with Singapore's Open Electricity Market (OEM) infrastructure.

- Programmed domain-specific data models conforming to the Market Support Services Licensee (MSSL) Electronic Business Transaction (EBT) standards.
- Engineered a multi-targeted architecture natively supporting a vast range of environments, including legacy .NET Framework (3.5 to 4.8.1), .NET Standard (2.0/2.1), and modern .NET platforms (.NET 5.0 through .NET 10.0).

- Developed strongly typed processing structures for high-volume XML messaging and compressed Base64/ZIP/CSV utility billing telemetry.
- Leveraged interfaces defined in the package to enable energy sector developers to build custom data streaming and parsing workflows.
- Maintained public developer documentation and API reference materials via GitHub Pages and SandCastle.

EMC NEMS Data Services Model Definitions | 2026
://jon-bong.github.io/emc-sewplus-nems-models/
C#, XSD, XML Serialisation, .NET Standard, .NET Core, NuGet, Sandcastle, GitHub

An open-source .NET class library that provides strongly typed model definitions for data services within the National Electricity Market of Singapore (NEMS) system managed by the Energy Market Company (EMC). The library acts as a robust data wrapper based on the official NEMS Data Services Specification.

- Strongly Typed Architecture: Designed strongly typed classes and enumerations to handle EMC Download Reports and Submission messages, ensuring type safety and reducing runtime errors.
- Flexible Serialisation: Implemented XML serialisation and deserialisation capabilities, allowing developers to process real-time market data (like Uniform Singapore Energy Price, demand, and reserve data) using individual types or broad report categories.
- Cross-Platform Compatibility: Engineered the package to support a massive ecosystem of target frameworks, including .NET Standard 2.0/2.1, .NET Core 3.0 up to .NET 10.0, and legacy .NET Framework 4.0+.
- Automated Documentation: Generated comprehensive API reference guides using Sandcastle to lower onboarding friction for external developers.

Continuous Learning and Development

AZ-900 Microsoft Azure Fundamentals | 2025
WSQ ICT-DIT-4020-1.1 Cloud Computing-4
Tertiary InfoTech Pte. Ltd.

AI-900 Azure AI Fundamentals | 2025
WSQ ICT-TEM-3034-1.1 Artificial Intelligence Application in Product Development-3
Tertiary InfoTech Pte. Ltd.

MECOMS 365 Functional Overview | 2024
Ferranti Pte Ltd

Low-Powered Wide-Area Networks | 2019
School of Engineering, Nanyang Polytechnic

System Operations on AWS | 2015
Amazon Web Services

Microsoft Official Course 20486B: Developing ASP.NET MVC 4 Web Applications | 2015
Microsoft Official Course 20480B: Programming in HTML5 with JavaScript and CSS3 | 2015
NetAssist Services Pte Ltd

Education

Monash University
Bachelor of Computing: System Development/Business Systems

Singapore Polytechnic
Diploma in Computer Information Systems

Anderson Junior College
General Certificate of Education Advanced (GCE A-Level)

Anderson Secondary School
General Certificate of Education Ordinary (GCE O-Level)